

Roll No.

TCE-401

**B. TECH. (CIVIL ENGINEERING)
(FOURTH SEMESTER)**

END SEMESTER EXAMINATION, June/July, 2022

HYDRAULICS AND HYDRAULIC MACHINES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive the condition for the rectangular channel of best section. (CO1)
- (b) Classify different types of flow with the help of specific energy diagram. (CO1)
- (c) An open channel of most economical section of trapezoidal shape is required to give a maximum discharge of $20.20 \text{ m}^3/\text{s}$ of water. The slope of the channel bottom is 1 in 2500. Taking Chezy's constant $C = 60$ in Chezy's equation, determine the dimensions of the cross-section. (CO1)

P. T. O.

2. (a) Derive the dynamic equation of gradually varied flow. (CO2)
- (b) Show that the brink depth is two-third of critical depth in an open rectangular channel. (CO2)
- (c) A rectangular channel 4 m wide conveys a discharge of $14 \text{ m}^3/\text{s}$ at a depth of 1.75 m. If the bed is raised by 0.20 m, will the water surface in the channel upstream be disturbed? Find the depth of flow over the raised bed and the fall in water surface. (CO2)
3. (a) What is meant by separation of boundary layer? Also discuss the various methods of controlling the boundary layer. (CO3)
- (b) Write short notes on displacement thickness, momentum thickness and energy thickness of boundary layer. (CO3)
- (c) A kite of dimensions $0.9 \text{ m} \times 0.9 \text{ m}$ and weighing 7.5 N is maintained in air at an angle of 10° to horizontal. The string attached to the kite makes an angle of 45° to the horizontal and at this position, the drag and lift coefficients are estimated to be 0.6 and 0.8 respectively. Make calculations for the wind velocity and tension in the string. Take density as 1.2 kg/m^3 . (CO3)
4. (a) Explain with sketches the working of a single stage centrifugal pump. (CO4)
- (b) Obtain an expression for the force exerted by a jet of water on a fixed inclined plate in the direction of the jet. (CO4)

(3)

- (c) A jet of water of 5 cm diameter impinges on a curved vane and is deflected through an angle of 175° . The vane moves in the same direction as that of the jet with a velocity of 35 m/s. If the water flow rate is 170 litres/second, determine the component of force on the vane in the direction of motion. (CO4)
5. (a) Write a short note on the factors influencing the performance of turbine. (CO5)
- (b) Briefly discuss the turbines according to the main direction of flow of water in the runner and give an example for each. (CO5)
- (c) The internal and external diameters of an outward flow reaction turbine are 2 m and 2.75 m respectively. The turbine is running at 250 r.p.m. and rate of flow of water through the turbine is $5 \text{ m}^3/\text{s}$. The width of runner is constant at inlet and outlet equal to 25 cm. The head at the turbine is 150 m. Neglect thickness of vanes and taking discharge radial at outlet, determine vane angles and flow velocity at inlet and outlet. (CO5)

Roll No.

TCE-402

B. TECH. (CE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

STRUCTURAL ANALYSIS—I

Time : Three Hours

Maximum Marks : 100

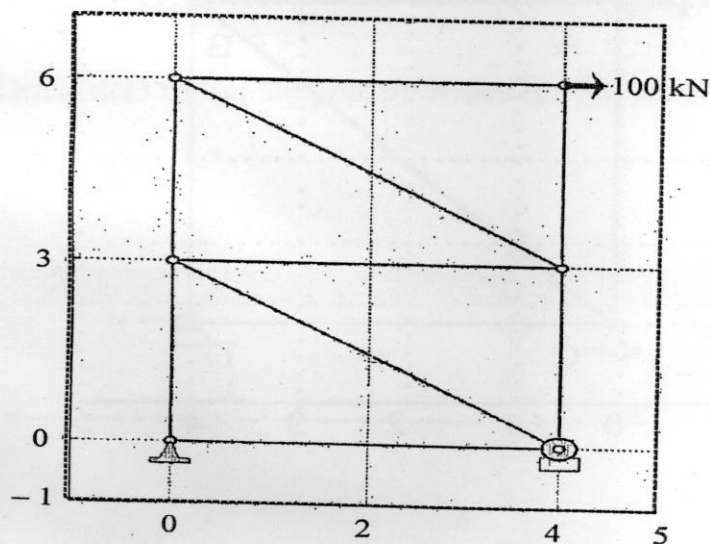
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

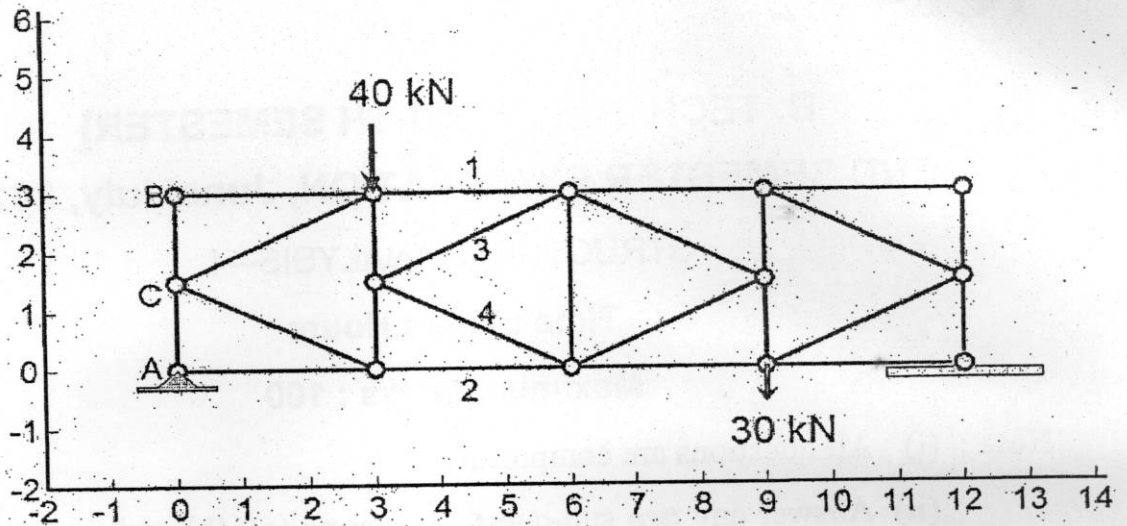
1. (a) Analyze the truss shown in Fig. below by method of joints to find all member forces. The dimensions shown in the scale are in metres :



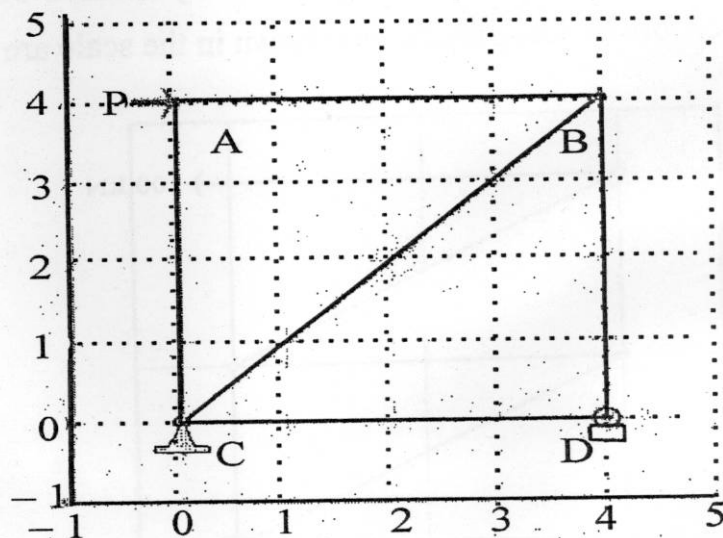
(CO1)

P. T. O.

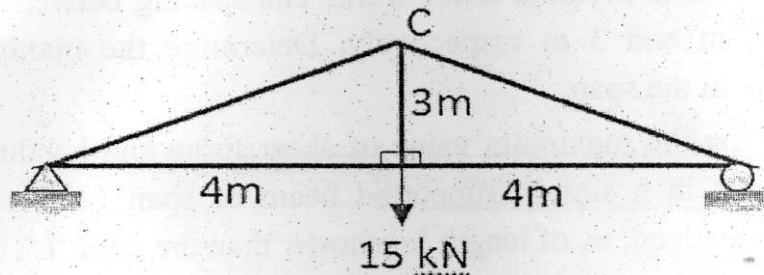
- (b) Determine the forces in the members 1, 2, 3 and 4 in the truss shown in Fig. below. Support at A is hinged and the support at the other end is roller. The dimensions on the scale are in metres : (CO1)



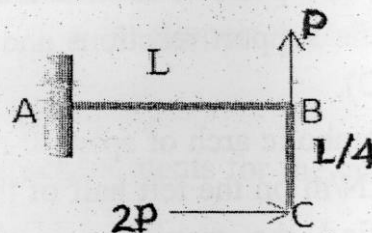
- (c) Using tension coefficient method, determine the forces in all the members of the truss shown in Fig. below. Support at joint C is hinge support, while that at D is Roller support. (CO1)



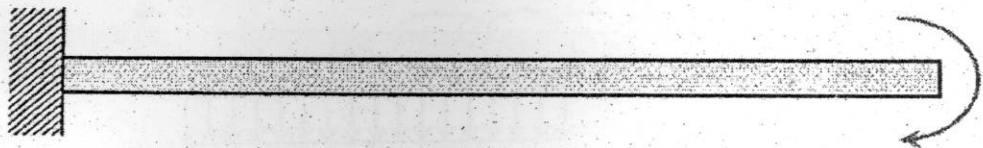
2. (a) Determine the vertical deflections at the joint C in the truss shown in Fig. below, using unit-load method : (CO2)



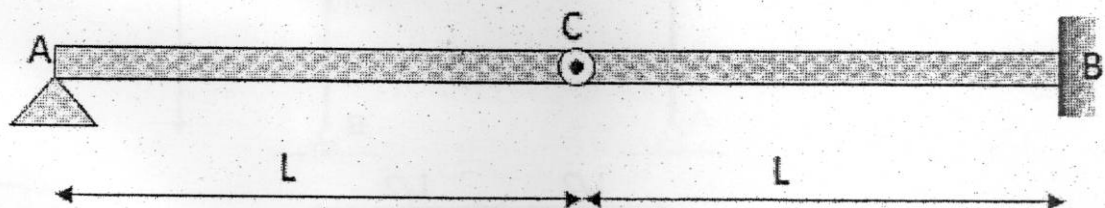
- (b) For the cantilever frame ABC shown in the Fig. below, find the horizontal deflection of the free-end C using any strain energy method. (CO2)



- (c) Determine the deflection and slope at the free end of the cantilever beam shown in Fig. below. Take the applied moment as M , the span of the beam as L , and flexural rigidity as EI . (CO2)

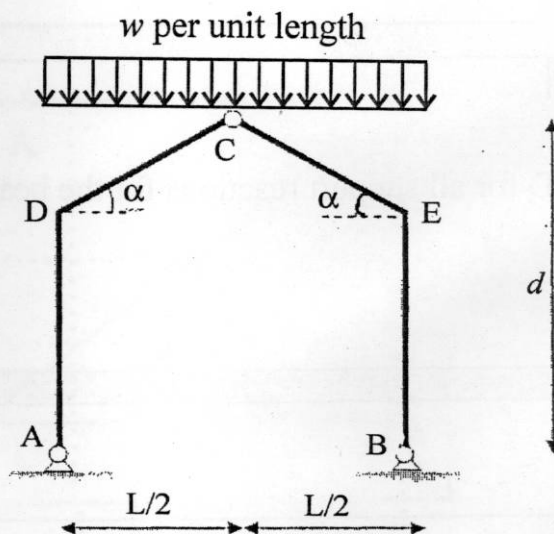


3. (a) Draw ILD for all support reactions for the beam shown in Fig. below : (CO3)



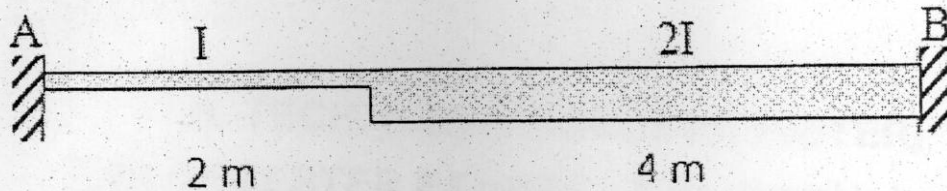
P. T. O.

- (b) A simply supported beam of span 20 m is traversed from left to right by the load system consisting of 5 kN, 20 kN, 2 kN and 2 kN. Of these loads, the 2 kN load is at the front. The spacing between the loads are 2 m, 3 m and 3 m respectively. Determine the maximum bending moment in the span. (CO3)
- (c) Determine the maximum value of shear force and bending moment at any point in a simply supported beam of span L , due to uniformly distributed load, w , of length ' c ' shorter than the span ' L '. (CO3)
4. (a) A three-hinged segmental arch ABC with a span ($AB =$) 50 m and a rise of 8 m at the crown 'C' is subjected to a 100 kN concentrated load (downward force) at a point 15 m measured horizontally from the right support B. Find the support reactions and sketch the bending moment distribution (BMD). (CO4)
- (b) A three-hinged parabolic arch of span 40 m and rise 10 m, is subjected to a UDL of 15 kN/m on the left half of the arch. Find reactions at the abutments and find the maximum positive and negative bending moments in the arch. (CO4)
- (c) A three-hinged gable frame is shown in Fig. below. Determine the support reactions. Sketch the bending moment and shear force diagram. (CO4)



5. (a) For the fixed beam shown in Fig. below, the support A rotates by 0.002 radian clockwise. Find the fixed end moment. Take $EI = 1013 \text{ N-mm}^2$.

(CO5)

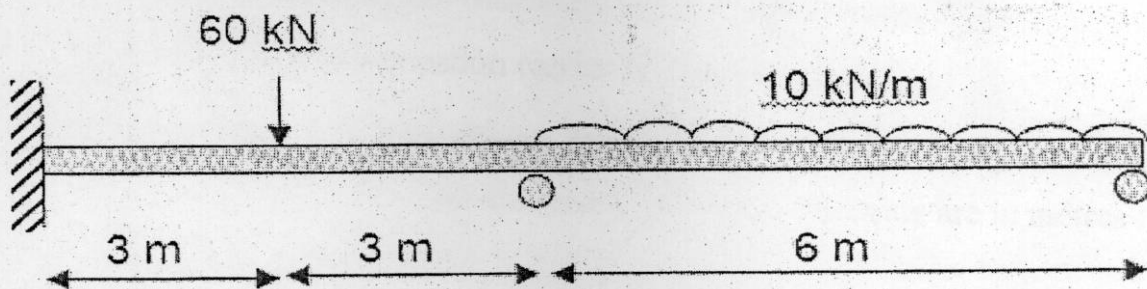


- (b) Draw the bending moment diagram of a propped cantilever beam AB subjected to UDL on its entire span. Take L as the span, w to be intensity of UDL. Consider support A as fixed and support B as roller.

(CO5)

- (c) Determine the support moments for the continuous beam shown in Fig. below :

(CO5)



Roll No.

TCE-403

**B. TECH. (CE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

ADVANCED SURVEYING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the classification of signals with the help of neat sketches.

(CO1)

(b) Describe the principle and objectives of triangulation. Differentiate between triangulation and trilateration.

(CO1)

(c) Explain the various criteria for selection of triangulation stations.

(CO1)

P. T. O.

2. (a) Determine the most probable values of the angles A, B and C from the following observation equations of equal weight : (CO2)

$$A = 30^\circ 24' 25.3''$$

$$B = 40^\circ 15' 14.2''$$

$$C = 70^\circ 39' 41.3''$$

The angles fulfill the conditions $A + B = C$.

- (b) Explain various laws of weights. (CO2)
- (c) The following are the measured values of the angle A with the corresponding weights : (CO2)

1	$51^\circ 20' 30''$	Weight 2
2	$51^\circ 20' 28''$	Weight 3
3	$51^\circ 20' 29''$	Weight 2

Determine :

- (i) The standard deviation.
 - (ii) The standard error of weighted mean.
 - (iii) The standard error of single observation of weight 3.
 - (iv) The probable error of the weighted mean.
3. (a) Two tangents intersect at a chainage of 1250.50 m having deflection angle of 60° . If the radius of the curve to be laid out is 375 m, calculate the Length of the curve, Tangent distance, Length of the long chord, Apex distance, Mid-ordinate, Degree of curve and Chainage of P.C. and P.T. (CO3)

- (b) Define transition curve. Describe the advantages of a transition curve. Explain requirements of a transition curve. Define super-elevation.

(CO3)

- (c) Explain field problems in setting out the circular curves. Explain various types of vertical curve with neat sketches. A vertical curve has an upgrade of 1.4% followed by a downgrade of 1%. The rate of change of grade is 0.12% per chain of 20 m. Calculate the length of vertical curve.

(CO3)

4. (a) Describe various stages of conducting field surveys for any civil engineering project.

(CO4)

- (b) Describe the purpose of field astronomy. Define the following terms :

(CO4)

(i) Celestial sphere

(ii) Equator

(iii) Latitude

(iv) Longitude

(v) Celestial horizon

(vi) Zenith

(vii) Nadir

(viii) Declination circle

- (c) Define hydro-graphic surveying. Explain the application and use of hydro-graphic surveying. Explain sounding in hydro-graphic surveying.

(CO4)

P. T. O.

5. (a) A line AB lies on a terrain having an average elevation of 400 m above mean sea level. It appears to be 8.72 cm on a photograph for which focal length is 24 cm. The same line measures 2.18 cm on a map which is to a scale of 1/40,000. Calculate the flying altitude of the aircraft above mean sea level when the photograph was taken. (CO5)
- (b) Explain Remote Sensing process in detail. (CO5)
- (c) Define Global Positioning System. Explain the principle and applications of GPS in different fields of Surveying. (CO5)

Roll No.

TCE-404

**B. TECH. (CIVIL ENGINEERING)
(FOURTH SEMESTER)**

END SEMESTER EXAMINATION, June/July, 2022

CONCRETE TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Enlist different types of cements. Discuss the properties and applications of two OPC based cements for concrete construction.

(CO1)

- (b) Describe unsoundness of cement. Explain the testing procedure to determine the unsoundness of cement with a neat sketch of the apparatus.

(CO1)

P. T. O.

- (c) Explain the role of strength of aggregate in quality and strength of concrete. Discuss briefly the two tests generally prescribed for determination of strength of aggregates. (CO1)
2. (a) Explain curing of concrete and its objectives. Enlist the various methods for curing and describe the membrane curing. (CO2)
- (b) Describe the basic requirements of formwork for concrete construction. (CO2)
- (c) Describe pumped concrete and enlist the precautions taken in pumping the concrete. (CO2)
3. (a) List the causes of lack of durability and explain how chloride-ion penetration affects the durability. (CO3)
- (b) Describe the destructive tests performed on hardened concrete and describe the split tension test on concrete cylinder and cube for determining the tensile strength of concrete. Enlist its merits of this indirect test. (CO3)
- (c) Describe the phenomenon of thermal expansion, transition zone of hardened concrete. (CO3)
4. (a) Explain the characteristics of aggregates required for mix design. (CO4)
- (b) Explain the major differences between ACI mix design method, British DoE method and IS-Concrete mix proportioning guidelines. (CO4)

(3)

- (c) Describe trial mixes and explain their role in achieving the objectives of mix design. (CO4)
- 5. (a) Explain the adverse effect of excessive use of admixtures. (CO5)
- (b) Explain G.G. blast furnace slag. (CO5)
- (c) Explain mineral or supplementary additives. Describe fly ash and silica fume. (CO5)

Roll No.

TCE-406

B. TECH. (CE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

ENGINEERING GEOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss about the internal structure of earth with neat diagrams. (CO1)
(b) Discuss about the major rock forming minerals. (CO1)
(c) Discuss the characteristics of minerals and their classification. (CO1)
2. (a) Discuss the engineering properties of rocks. Also mention *five* examples each of igneous, sedimentary and metamorphic rocks. (CO2)
(b) Discuss about the origin of igneous, sedimentary and metamorphic rocks. (CO2)
(c) Explain metamorphism. Also discuss about the classification of metamorphic rocks. (CO2)

P. T. O.

(2)

3. (a) Define a landslide. Also discuss the causes of the landslides. (CO3)
(b) Define unconformity in rocks and discuss its classification. (CO3)
(c) Discuss about the causes and classification of joint in rocks. (CO3)
4. (a) Define intensity of an earthquake. Also discuss the modified mercalli intensity scale. (CO4)
(b) Discuss the zonal classification for seismic zones in India. (CO4)
(c) Define magnitude of an earthquake. Also discuss the richter scale for measurement of magnitude. (CO4)
5. (a) Discuss about the geological classifications for site selection of dams. (CO5)
(b) Discuss about reservoir induces seismicity and its causes. (CO5)
(c) Define a reservoir and discuss its types. (CO5)